

Sharp linear hypercontractivity for ultraspherical measures

Abstract

The linear hypercontractive estimate of Ivanisvili, Lindemberger, Müller and Schmuckenschläger on the circle, proved in their work for exponents $6 \leq p \leq q$, holds for every admissible pair. For $m \geq -1$ and $q \geq p > 0$ with $p + m > 0$,

$$\|1 + rbz\|_{L^q(\nu_m)} \leq \|1 + bz\|_{L^p(\nu_m)}$$

for all real b , if and only if $|r| \leq \sqrt{(p+m)/(q+m)}$. At $m = -1$ the restriction $p + m > 0$ is Bonami's condition $p > 1$. The same threshold is sharp for the corresponding linear estimate on every sphere.

1 Statement

For $m > -1$ let ν_m be the ultraspherical probability measure on the unit circle,

$$d\nu_m(e^{i\theta}) = c_m |\sin \theta|^m d\theta, \quad c_m = \frac{\Gamma(m/2 + 1)}{2\sqrt{\pi} \Gamma(m/2 + 1/2)},$$

the integral running over $[0, 2\pi]$, and let $\nu_{-1} = \frac{1}{2}(\delta_1 + \delta_{-1})$. The constant c_m is chosen so that ν_m is a probability measure. Indeed,

$$\int_0^{\pi/2} \sin^m \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{1}{2}\right) = \frac{\sqrt{\pi} \Gamma(m/2 + 1/2)}{2 \Gamma(m/2 + 1)},$$

and four copies of this integral give

$$\int_0^{2\pi} |\sin \theta|^m d\theta = 2\sqrt{\pi} \frac{\Gamma(m/2 + 1/2)}{\Gamma(m/2 + 1)}.$$

The estimate under consideration is

$$\|1 + rbz\|_{L^q(\nu_m)} \leq \|1 + bz\|_{L^p(\nu_m)} \quad \text{for every } b \in \mathbb{R}. \quad (1)$$

Theorem 1. *Let $m \geq -1$ and $q \geq p > 0$ with $p + m > 0$. Then (1) holds for every real b if and only if*

$$|r| \leq \sqrt{\frac{p+m}{q+m}}.$$

At $m = -1$, the hypothesis $p + m > 0$ is exactly $p > 1$.

The case $m = -1$ is Bonami's theorem, and the case $m = 0$ is Weissler's theorem on the circle. Theorem 1 removes the restriction $6 \leq p \leq q$ from Theorem 3.1 of Ivanisvili–Lindemberger–Müller–Schmuckenschläger, and it extends their Theorem 2.1 from the exponents $0 < p \leq 2 = q$ on the sphere to every pair $q \geq p > 0$.

Theorem 2. Let $n \geq 2$, let σ be the uniform probability measure on $\mathbb{S}^{n-1}$, and let $q \geq p > 0$. For every $x \in \mathbb{R}^n$ and every real a ,

$$\|x + ary\|_{L^q(\mathbb{S}^{n-1}, \sigma)} \leq \|x + ay\|_{L^p(\mathbb{S}^{n-1}, \sigma)}$$

whenever $|r| \leq \sqrt{(p+n-2)/(q+n-2)}$. The constant is sharp.

Throughout, $m = 2\lambda$ with $\lambda > -1/2$, and μ_λ denotes the law of $\operatorname{Re} z$ under ν_m . Explicitly, μ_λ is the probability measure on $[-1, 1]$ with density

$$\frac{\Gamma(\lambda + 1)}{\sqrt{\pi} \Gamma(\lambda + 1/2)} (1 - t^2)^{\lambda - 1/2}.$$

Indeed, $\cos \theta = t$ has two preimages in $[0, 2\pi)$, each with Jacobian $1/\sqrt{1 - t^2}$, so the density of the push-forward is

$$2c_{2\lambda}(1 - t^2)^{\lambda - 1/2},$$

and $2c_{2\lambda}$ is the constant displayed above. For $c \geq 0$ write

$$F_c(t) = 1 + 2ct + c^2$$

and

$$N_\alpha(c) = \left(\int_{-1}^1 F_c(t)^\alpha d\mu_\lambda(t) \right)^{1/\alpha},$$

for every $\alpha > 0$ with $\alpha + \lambda > 0$. Since ν_m is invariant under $z \mapsto -z$,

$$\|1 + bz\|_{L^{2\alpha}(\nu_m)} = N_\alpha(|b|)^{1/2}.$$

2 Necessity

Proposition 3. If $m > -1$ and $p + m > 0$, $q + m > 0$, and if (1) holds for every small real b , then $|r| \leq \sqrt{(p+m)/(q+m)}$.

Proof. The identity $\cos^2 \theta = 1 - \sin^2 \theta$ gives

$$\int (\operatorname{Re} z)^2 d\nu_m = 1 - \frac{c_m}{c_{m+2}}.$$

The recurrence $\Gamma(w+1) = w\Gamma(w)$ gives

$$\frac{c_m}{c_{m+2}} = \frac{\Gamma(m/2 + 1)}{\Gamma(m/2 + 2)} \cdot \frac{\Gamma(m/2 + 3/2)}{\Gamma(m/2 + 1/2)} = \frac{m/2 + 1/2}{m/2 + 1} = \frac{m+1}{m+2},$$

so the second moment equals $1 - (m+1)/(m+2) = 1/(m+2)$.

For $|b| < 1/2$ one has $1 + 2b \operatorname{Re} z + b^2 \geq (1 - |b|)^2 > 1/4$. Taylor expansion with a uniform remainder therefore yields

$$\begin{aligned} \int (1 + 2b \operatorname{Re} z + b^2)^{p/2} d\nu_m &= 1 + \frac{p}{2} b^2 + \frac{p(p-2)}{2(m+2)} b^2 + O(b^3) \\ &= 1 + \frac{pb^2}{2} \cdot \frac{p+m}{m+2} + O(b^3). \end{aligned}$$

The odd contributions in b integrate to zero by the symmetry $z \mapsto -z$. Raising to the power $1/p$ gives

$$\|1 + bz\|_{L^p(\nu_m)} = 1 + \frac{b^2}{2} \cdot \frac{p+m}{m+2} + O(b^3).$$

The same expansion at exponent q , with b replaced by rb , gives

$$\|1 + rbz\|_{L^q(\nu_m)} = 1 + \frac{r^2 b^2}{2} \cdot \frac{q+m}{m+2} + O(b^3).$$

Subtract the constant term 1, divide by b^2 , and let $b \rightarrow 0$. Inequality (1) forces

$$r^2 \frac{q+m}{m+2} \leq \frac{p+m}{m+2}.$$

□

The same expansion at $m = -1$, where $\int (\operatorname{Re} z)^2 d\nu_{-1} = 1$, gives the necessity of $|r| \leq \sqrt{(p-1)/(q-1)}$ for $p, q > 1$.

3 The hypergeometric representation

For $0 \leq c < 1$ and every real α ,

$$\int F_c^\alpha d\mu_\lambda = {}_2F_1(-\alpha, -\alpha - \lambda; \lambda + 1; c^2). \quad (2)$$

The identity $F_c = c^2 F_{1/c}$ gives, for every $c > 0$,

$$N_\alpha(c) = c^2 N_\alpha(1/c). \quad (3)$$

To prove (2), set $z = c^2$ and $y(z) = \int F_{\sqrt{z}}^\alpha d\mu_\lambda$. The function

$$Y(z) = {}_2F_1(-\alpha, -\alpha - \lambda; \lambda + 1; z)$$

is the solution of

$$z(1-z)Y'' + [\lambda + 1 - (1 - 2\alpha - \lambda)z]Y' - \alpha(\alpha + \lambda)Y = 0 \quad (4)$$

which is analytic at the origin and satisfies $Y(0) = 1$. The constant term of y is 1. The coefficient of z is computed from

$$F_c^\alpha = 1 + \alpha(b^2 + 2bt) + \alpha(\alpha - 1)2b^2t^2 + O(b^3),$$

with $b = \sqrt{z}$. The remainder $O(b^3)$ is $O(z^{3/2})$, hence $o(z)$. Together with $\int t d\mu_\lambda = 0$ and

$$\int t^2 d\mu_\lambda = \frac{1}{2(\lambda + 1)}.$$

(The second moment is the case $m = 2\lambda$ of the computation in Proposition 3.) Integrating and using $\int t d\mu_\lambda = 0$ produces

$$y(z) = 1 + \alpha z + \frac{\alpha(\alpha - 1)}{\lambda + 1} z + o(z) = 1 + \frac{\alpha(\alpha + \lambda)}{\lambda + 1} z + o(z),$$

so

$$y'(0) = \frac{\alpha(\alpha + \lambda)}{\lambda + 1}.$$

The derivative of ${}_2F_1(a, b; c_*; z)$ at the origin equals ab/c_* . With $a = -\alpha$, $b = -\alpha - \lambda$ and $c_* = \lambda + 1$, this is the same value. The map $c \mapsto \int F_c^\alpha d\mu_\lambda$ is analytic for $|c| < 1$, because $F_c \geq (1 - |c|)^2 > 0$, and it is even by the substitution $t \mapsto -t$. An even analytic function of c is analytic in c^2 , so y is analytic on $[0, 1)$. The hypergeometric series is the unique analytic solution of (4) with value 1 at the origin: the constant term of the differential equation forces the coefficient of z , and each higher coefficient is then determined by the recurrence. Thus it is enough to show that y satisfies (4) on $(0, 1)$.

Lemma 4. *Let $c \in \mathbb{R}$ and let $F = F_c$. On $(-1, 1)$,*

$$\frac{d}{dt} \left(F^{\alpha-1} (1 - t^2)^{\lambda+1/2} \right) = -Q F^{\alpha-2} (1 - t^2)^{\lambda-1/2},$$

where

$$Q = (2\lambda + 1)(1 + c^2)t + 2c(\alpha + 2\lambda)t^2 + 2c(1 - \alpha).$$

Proof. Differentiate the product. With $F' = 2c$,

$$\begin{aligned} \frac{d}{dt} \left(F^{\alpha-1} (1 - t^2)^{\lambda+1/2} \right) &= 2c(\alpha - 1)F^{\alpha-2}(1 - t^2)^{\lambda+1/2} - t(2\lambda + 1)F^{\alpha-1}(1 - t^2)^{\lambda-1/2} \\ &= F^{\alpha-2}(1 - t^2)^{\lambda-1/2} \left(2c(\alpha - 1)(1 - t^2) - t(2\lambda + 1)F \right). \end{aligned}$$

The expression in parentheses equals $-Q$, because

$$\begin{aligned} t(2\lambda + 1)F - 2c(\alpha - 1)(1 - t^2) &= (2\lambda + 1)(1 + c^2)t + 2c(2\lambda + 1)t^2 + 2c(\alpha - 1)t^2 - 2c(\alpha - 1) \\ &= (2\lambda + 1)(1 + c^2)t + 2c(\alpha + 2\lambda)t^2 + 2c(1 - \alpha). \end{aligned}$$

□

For $c \in (0, 1)$ one has $F \geq (1 - c)^2 > 0$, so $F^{\alpha-1}$ is bounded on $[-1, 1]$. The boundary factor $(1 - t^2)^{\lambda+1/2}$ vanishes at $t = \pm 1$ because $\lambda > -1/2$. The derivative on the left of Lemma 4 is integrable on $(-1, 1)$, since near each endpoint it is comparable to $(1 - t^2)^{\lambda-1/2}$ and $\lambda - 1/2 > -1$. Consequently

$$\int Q F^{\alpha-2} d\mu_\lambda = 0.$$

Now let $I(c) = \int F_c^\alpha d\mu_\lambda$, so that $y(z) = I(\sqrt{z})$. Then

$$y_z = \frac{I_c}{2c}, \quad y_{zz} = \frac{I_{cc}}{4c^2} - \frac{I_c}{4c^3}.$$

Substituting $z = c^2$ into the differential expression (4) and clearing the denominator $4c^3$ produces, under the integral sign, the integrand identity

$$\begin{aligned} z(1 - z)\partial_{zz}(F^\alpha) + [\lambda + 1 - (1 - 2\alpha - \lambda)z]\partial_z(F^\alpha) - \alpha(\alpha + \lambda)F^\alpha \\ = -\frac{\alpha(c^2 - 1)}{2c} Q F^{\alpha-2}. \end{aligned} \tag{5}$$

(The two sides are rational functions of c and t , once $F^{\alpha-2}$ is divided out, and the resulting polynomial identity is the expansion of Lemma 4 against the chain rule; it is recorded in the appendix.) Integrating (5) and using $\int Q F^{\alpha-2} d\mu_\lambda = 0$ shows that y satisfies (4). This proves (2).

For $\alpha > 0$ and $z \in (0, 1)$ define

$$\tau_\alpha(z) = \frac{1}{\alpha} \partial_z \log y(\alpha, z) = \frac{\partial_z y(\alpha, z)}{\alpha y(\alpha, z)},$$

where $y(\alpha, z)$ denotes the right-hand side of (2). Then τ_α extends continuously to $z = 0$ by

$$\tau_\alpha(0) = \frac{\alpha + \lambda}{\lambda + 1},$$

and the substitution $\partial_z \log y = \alpha \tau_\alpha$ turns (4) into the Riccati equation

$$z(1 - z)\tau' = (\alpha + \lambda) - [\lambda + 1 - (1 - 2\alpha - \lambda)z]\tau - z(1 - z)\alpha\tau^2. \quad (6)$$

In particular,

$$\frac{1}{\alpha} \log y(\alpha, z) = \int_0^z \tau_\alpha(t) dt. \quad (7)$$

Lemma 5. *If $\alpha > 0$ and $\alpha + \lambda > 0$, then $\tau_\alpha(z) > 0$ for every $z \in [0, 1)$.*

Proof. The value at the origin is positive. If $z \in (0, 1)$ were a first zero, (6) at $\tau = 0$ would give $z(1 - z)\tau' = \alpha + \lambda > 0$, so $\tau' > 0$, which is incompatible with reaching zero from above. $\square$

4 An upper bound for the logarithmic derivative

Lemma 6. *For every $\lambda > -1/2$, every $\alpha > 0$ and every $z \in (0, 1)$,*

$$\tau_\alpha(z) < \frac{1}{\sqrt{z}(1 + \sqrt{z})}.$$

In particular $\tau_\alpha(z) < 1/(2z)$.

Proof. Write $\sigma(z) = 1/(\sqrt{z}(1 + \sqrt{z}))$ and $s = \sqrt{z}$, so that $\sigma(z) = 1/(s(s + 1))$. As $z \downarrow 0$ one has $\sigma(z) \rightarrow \infty$, while $\tau_\alpha(z)$ tends to the finite value $(\alpha + \lambda)/(\lambda + 1)$. Thus $\tau_\alpha < \sigma$ on a right neighbourhood of the origin.

Set $t = 1/(s(s + 1))$ and substitute $\tau = t$, $z = s^2$ into the right-hand side of (6). The quadratic term contributes

$$z(1 - z)\alpha\tau^2 \cdot s(s + 1) = \alpha s^2(1 - s^2) \cdot \frac{1}{s(s + 1)} = \alpha s(1 - s),$$

where $1 - s^2 = (1 - s)(1 + s)$ and $(1 + s)/(s + 1) = 1$. Multiplying the whole right-hand side by $s(s + 1)$ therefore gives

$$\begin{aligned} & (\alpha + \lambda)s(s + 1) - [\lambda + 1 - (1 - 2\alpha - \lambda)s^2] - \alpha s(1 - s) \\ &= s^2 + \lambda s - (\lambda + 1) = (s - 1)(s + \lambda + 1). \end{aligned}$$

(The coefficient of s^2 is $(\alpha + \lambda) + (1 - 2\alpha - \lambda) + \alpha = 1$, and the coefficient of s is $(\alpha + \lambda) - \alpha = \lambda$.) Therefore, at every point where $\tau_\alpha = \sigma$,

$$\tau'_\alpha = \frac{(s - 1)(s + \lambda + 1)}{s(s + 1) \cdot s^2(1 - s^2)} = -\frac{s + \lambda + 1}{s^3(s + 1)^2},$$

where $1 - s^2 = (1 - s)(1 + s)$ and $s - 1 = -(1 - s)$ were used. On the other hand,

$$\frac{d}{ds} \frac{1}{s(s+1)} = -\frac{1+2s}{s^2(s+1)^2}, \quad \frac{ds}{dz} = \frac{1}{2s},$$

so

$$\sigma'(z) = -\frac{1+2s}{2s^3(s+1)^2}.$$

Subtracting these evaluations yields

$$\tau'_\alpha - \sigma' = \frac{-2(s + \lambda + 1) + (1 + 2s)}{2s^3(s+1)^2} = -\frac{2\lambda + 1}{2s^3(s+1)^2} < 0,$$

because $\lambda > -1/2$. At a first zero of $\sigma - \tau_\alpha$ in $(0, 1)$ one would have $\tau'_\alpha \geq \sigma'$. This contradiction keeps $\tau_\alpha < \sigma$ on the whole interval.

Finally, $z + \sqrt{z} \geq 2z$ for $z \in (0, 1]$, so $\sigma(z) \leq 1/(2z)$, and the inequality is strict for $z \in (0, 1)$. $\square$

Lemma 7. *For every $\alpha > 0$ with $\alpha + \lambda > 0$, the map $c \mapsto N_\alpha(c)$ is continuous and strictly increasing on $[0, \infty)$.*

Proof. For $c \in (0, 1)$, (2) and the chain rule give

$$\frac{N'_\alpha(c)}{N_\alpha(c)} = 2c\tau_\alpha(c^2).$$

Lemma 5 makes the right-hand side positive. For $c > 1$, (3) gives

$$\frac{d}{dc} \log N_\alpha(c) = \frac{2}{c} - \frac{1}{c^2} \cdot \frac{N'_\alpha(1/c)}{N_\alpha(1/c)} = \frac{2}{c} - \frac{2}{c^3} \tau_\alpha(c^{-2}).$$

Lemma 6 yields $\tau_\alpha(c^{-2}) < c^2/2$, so the derivative is positive. Continuity on $[0, \infty)$ holds because $0 \leq F_c \leq (1+c)^2$ and $\alpha > 0$. $\square$

Lemma 7 removes the restriction to the critical multiplier. The law of z is symmetric, so b may be replaced by $|b|$. For $0 \leq r \leq r_*$ and $c \geq 0$,

$$N_{q/2}(rc) \leq N_{q/2}(r_*c).$$

Thus (1) holds for every r with $|r| \leq \sqrt{(p+m)/(q+m)}$ as soon as it holds at the positive endpoint.

5 Scales at most one

Set $\gamma = p/2$ and $\delta = q/2$. The hypotheses $q \geq p > 0$ and $p+m > 0$ become $0 < \gamma \leq \delta$ and $\gamma + \lambda > 0$. Set

$$\kappa = \frac{\gamma + \lambda}{\delta + \lambda} \in (0, 1].$$

Inequality (1) at the critical multiplier is equivalent to

$$N_\gamma(c) \geq N_\delta(c\sqrt{\kappa}) \quad \text{for every } c \geq 0. \quad (8)$$

Lemma 8. *For every $z \in (0, 1)$,*

$$\tau_\gamma(z) > \kappa \tau_\delta(\kappa z)$$

whenever $\delta > \gamma$, and therefore $N_\gamma(\sqrt{z}) \geq N_\delta(\sqrt{\kappa z})$. In particular, (8) holds for every $c \in [0, 1]$.

Proof. The case $\gamma = \delta$ is the identity $\kappa = 1$, so assume $\delta > \gamma$. Write $\psi(z) = \kappa \tau_\delta(\kappa z)$. Then $\tau_\gamma(0) = \psi(0) = (\gamma + \lambda)/(\lambda + 1)$. Differentiate (6). The derivative of the left side is $(1 - 2z)\tau' + z(1 - z)\tau''$, which equals $\tau'(0)$ at $z = 0$. At $z = 0$ and $\tau = t_0 = (\alpha + \lambda)/(\lambda + 1)$ both sides of (6) vanish. Differentiating the right side, the product rule contributes $(1 - 2\alpha - \lambda)\tau$ from the coefficient of τ and $-\alpha\tau^2$ from $z(1 - z)\alpha\tau^2$, together with the terms that still contain τ' . At the origin this equals

$$(1 - 2\alpha - \lambda)t_0 - (\lambda + 1)\tau'(0) - \alpha t_0^2.$$

Solving for the derivative gives

$$\tau'_\alpha(0) = \frac{t_0}{\lambda + 2}(1 - 2\alpha - \lambda - \alpha t_0). \quad (9)$$

The chain rule yields $\psi'(0) = \kappa^2 \tau'_\delta(0)$. Since $\kappa t_\delta = t_\gamma$, with $t_\alpha = (\alpha + \lambda)/(\lambda + 1)$,

$$\tau'_\gamma(0) - \psi'(0) = \frac{t_\gamma}{\lambda + 2} \left((1 - 2\gamma - \lambda - \gamma t_\gamma) - \kappa(1 - 2\delta - \lambda - \delta t_\delta) \right).$$

The expression in parentheses equals

$$\frac{\delta - \gamma}{(\delta + \lambda)(\lambda + 1)} (\gamma\delta + \lambda(\gamma + \delta) + 2\lambda^2 + 2\lambda + 1).$$

(The verification is the expansion in the appendix.) The quadratic form factors as

$$\gamma\delta + \lambda(\gamma + \delta) + 2\lambda^2 + 2\lambda + 1 = (\gamma + \lambda)(\delta + \lambda) + (\lambda + 1)^2 > 0.$$

Hence $\tau'_\gamma(0) - \psi'(0) > 0$. Combined with $\tau_\gamma(0) = \psi(0)$, this forces $\tau_\gamma > \psi$ on a right neighbourhood of the origin.

Suppose there exists a point of $(0, 1)$ at which $\tau_\gamma = \psi$. Let z_* be the infimum of such points. Then $z_* > 0$, $\tau_\gamma(z_*) = \psi(z_*)$, and $\tau_\gamma - \psi \geq 0$ on $[0, z_*]$, so the derivative of the difference at z_* is at most zero. It remains to rule that out.

Let $z \in (0, 1)$ satisfy $\tau_\gamma(z) = \psi(z) = t$. The equation (6) at exponent γ reads

$$z(1 - z)\tau'_\gamma = N_\gamma(t),$$

where

$$N_\gamma(t) = (\gamma + \lambda) - [\lambda + 1 - (1 - 2\gamma - \lambda)z]t - z(1 - z)\gamma t^2.$$

For the comparison function, set $u = \kappa z$ and $\tau_\delta(u) = \psi(z)/\kappa$. Then $\psi'(z) = \kappa^2 d\tau_\delta/du$. Substitute into (6) at exponent δ , multiply through by κ^2 , and use $\kappa^2(\delta + \lambda) = \kappa(\gamma + \lambda)$. Dividing by $\kappa > 0$ produces

$$z(1 - \kappa z)\psi' = N_\psi(t), \quad (10)$$

where

$$N_\psi(t) = (\gamma + \lambda) - [\lambda + 1 - (1 - 2\delta - \lambda)\kappa z]t - z(1 - \kappa z)\delta t^2.$$

Therefore

$$\tau'_\gamma(z) - \psi'(z) = \frac{N_\gamma(t)(1 - \kappa z) - N_\psi(t)(1 - z)}{z(1 - z)(1 - \kappa z)}.$$

The denominator is positive. The numerator equals

$$\frac{z(\delta - \gamma)}{\delta + \lambda} Q(t),$$

where

$$Q(t) = A_Q t^2 - 2z(\gamma + \lambda)t + (\gamma + \lambda), \quad A_Q = (1 - z)(\delta + \lambda - z(\gamma + \lambda)).$$

This polynomial identity is checked coefficient-wise in the appendix. The prefactor $z(\delta - \gamma)/(\delta + \lambda)$ is positive, so the sign of $\tau'_\gamma - \psi'$ is the sign of $Q(t)$.

Since $\delta \geq \gamma$ and $z \in (0, 1)$,

$$A_Q - (1 - z)^2(\gamma + \lambda) = (1 - z)(\delta - \gamma) \geq 0,$$

and hence

$$Q(t) \geq (\gamma + \lambda)((1 - z)^2 t^2 - 2zt + 1).$$

For $z \in (0, 1/2]$ the discriminant of the quadratic in t is $4(2z - 1) \leq 0$, so the quadratic is positive for every finite t except at the single point $z = 1/2, t = 2$, where it vanishes. For $z \in (1/2, 1)$ the roots are

$$\frac{1}{z \pm \sqrt{2z - 1}},$$

because

$$(z - \sqrt{2z - 1})(z + \sqrt{2z - 1}) = (1 - z)^2.$$

The quadratic is then nonnegative for $0 \leq t \leq 1/(z + \sqrt{2z - 1})$. Lemma 6 supplies the stricter bound

$$t = \tau_\gamma(z) < \frac{1}{z + \sqrt{z}} \leq \frac{1}{z + \sqrt{2z - 1}}$$

on $[1/2, 1)$, the comparison of the denominators being $\sqrt{z} \geq \sqrt{2z - 1}$. At $z = 1/2$ the same bound gives $\tau_\gamma(1/2) < 1/(1/2 + \sqrt{1/2}) < 2$. Thus $Q(\tau_\gamma(z)) > 0$, and $\tau'_\gamma > \psi'$ at every contact point in $(0, 1)$.

The graphs cannot meet in $(0, 1)$. Therefore $\tau_\gamma(z) > \psi(z)$ for every $z \in (0, 1)$. Integrating and applying (7) gives

$$\frac{1}{\gamma} \log y(\gamma, z) \geq \frac{1}{\delta} \log y(\delta, \kappa z),$$

which is $N_\gamma(\sqrt{z}) \geq N_\delta(\sqrt{\kappa z})$. The endpoint $c = 1$ follows by continuity of N_α . $\square$

6 Scales larger than one

Lemma 9. *For each fixed $\alpha > 0$ with $\alpha + \lambda > 0$, the map $a \mapsto N_\alpha(a)/a$ is strictly decreasing on $(0, 1]$.*

Proof. For $a \in (0, 1)$,

$$\frac{d}{da} \log \frac{N_\alpha(a)}{a} = 2a \tau_\alpha(a^2) - \frac{1}{a}.$$

Lemma 6 yields $2a \tau_\alpha(a^2) < 2a \cdot (1/(2a^2)) = 1/a$, so the derivative is negative. If $0 < a < 1$, the same derivative on $(a, 1)$ gives $N_\alpha(a)/a > N_\alpha(1)/1$. Thus the map is strictly decreasing on $(0, 1]$. $\square$

In particular, $0 < b < a \leq 1$ implies $N_\alpha(b)/b > N_\alpha(a)/a$.

Proposition 10. *Inequality (8) holds for every $c > 1$.*

Proof. Write $\rho(c) = N_\gamma(c)/N_\delta(c\sqrt{\kappa})$. Lemma 8 at scale $1/c \in (0, 1)$ gives

$$N_\gamma(1/c) \geq N_\delta((1/c)\sqrt{\kappa}).$$

Suppose first that $c\sqrt{\kappa} > 1$, and set $a = 1/(c\sqrt{\kappa}) \in (0, 1)$. Then $(1/c)\sqrt{\kappa} = \kappa a$. The scaling relation (3) gives

$$\rho(c) = \frac{N_\gamma(1/c)}{\kappa N_\delta(a)} = \frac{N_\gamma(1/c)}{N_\delta(\kappa a)} \cdot \frac{N_\delta(\kappa a)}{\kappa N_\delta(a)}.$$

The first factor is at least 1 by Lemma 8. If $\kappa < 1$, then $0 < \kappa a < a < 1$, so Lemma 9 makes the second factor strictly larger than 1. If $\kappa = 1$, both factors equal 1. Thus $\rho(c) \geq 1$.

Suppose instead that $c > 1$ and $c\sqrt{\kappa} \leq 1$. Then necessarily $\kappa < 1$. Set $x = c\sqrt{\kappa} \in (\sqrt{\kappa}, 1]$ and $y = \kappa/x$. The inequality $x > \sqrt{\kappa}$ forces $y < x$, and both points lie in $(0, 1]$. Also $y = (1/c)\sqrt{\kappa}$ and $c^2 = x^2/\kappa$. Hence

$$\rho(c) = \frac{c^2 N_\gamma(1/c)}{N_\delta(x)} \geq \frac{x^2}{\kappa} \cdot \frac{N_\delta(y)}{N_\delta(x)},$$

where the inequality is Lemma 8. The relation $\kappa = xy$ rewrites the desired bound $x^2 N_\delta(y) \geq \kappa N_\delta(x)$ as

$$\frac{N_\delta(y)}{y} \geq \frac{N_\delta(x)}{x},$$

which is Lemma 9. Therefore $\rho(c) \geq 1$. □

Lemmas 8 and 10 are (8) for every $c \geq 0$. By the reduction of Section 7 below, this is Theorem 1 for $m > -1$.

7 The endpoint and the sphere

For $m > -1$, the identity $\|1 + bz\|_{L^{2\alpha}(\nu_m)} = N_\alpha(|b|)^{1/2}$ converts (8) into (1) at

$$r = \sqrt{\kappa} = \sqrt{\frac{p+m}{q+m}}.$$

Lemma 7 extends the inequality to every smaller multiplier, and Proposition 3 gives the matching necessity.

If $m = -1$ and $p > 1$, then $p + m' > 0$ for every $m' \in (-1, 0)$. The case $m > -1$ of the theorem, already proved, gives (1) for $\nu_{m'}$ at the multiplier $\sqrt{(p+m')/(q+m')}$. As $m' \downarrow -1$,

$$\int (\operatorname{Re} z)^2 d\nu_{m'} = \frac{1}{m' + 2} \rightarrow 1.$$

Every weak-star limit point ν satisfies $\int (\operatorname{Re} z)^2 d\nu = 1$. Then $\int (1 - (\operatorname{Re} z)^2) d\nu = 0$. The integrand is continuous and nonnegative, so ν is supported on $\{\pm 1\}$. Each $\nu_{m'}$ is invariant under $z \mapsto -z$, so the same is true of ν , and therefore $\nu = \nu_{-1}$. The circle is compact, so $\nu_{m'}$ converges weak-star to ν_{-1} . Fix $b \in \mathbb{R}$ and set $r(m') = \sqrt{(p+m')/(q+m')}$. Then $r(m')$ tends to $r_* = \sqrt{(p-1)/(q-1)}$, and $z \mapsto |1 + r(m')bz|^q$ converges uniformly to $z \mapsto |1 + r_*bz|^q$. The function $z \mapsto |1 + bz|^p$ is continuous. Weak-star convergence preserves both integrals, so the norms pass to the limit and (1) holds for ν_{-1} at the multiplier r_* . The expansion recorded after Proposition 3 gives the matching necessity.

Proof of Theorem 2. If $x = 0$, then $|ay|$ is the constant $|a|$ for every $y \in \mathbb{S}^{n-1}$, and the asserted bound reduces to $|r| \leq 1$. This holds because $q \geq p$. Both sides are homogeneous of degree one in (x, a) , so one may assume $|x| = 1$. For $y \in \mathbb{S}^{n-1}$,

$$|x + ay|^2 = 1 + 2a\langle x, y \rangle + a^2.$$

The set $\{y \in \mathbb{S}^{n-1} : \langle x, y \rangle \in dt\}$ has area proportional to $(1 - t^2)^{(n-3)/2} dt$. With $\lambda = (n - 2)/2$, the exponent $(n - 3)/2$ equals $\lambda - 1/2$, so the law of $\langle x, y \rangle$ is μ_λ , and the corresponding circle weight is $m = n - 2$. Symmetry in t gives

$$\|x + ay\|_{L^p(\sigma)} = N_{p/2}(|a|)^{1/2},$$

and the same identity with a replaced by ar at the exponent q . Theorem 1 at weight $m = n - 2$ is therefore the spherical estimate. Sharpness is Proposition 3 at this weight. $\square$

Appendix: the polynomial identities

The integrand form of the hypergeometric operator

Let $z = c^2$, $F = 1 + 2ct + c^2$ and $\partial_c F = 2(c + t)$. Write $L[F^\alpha]$ for the left-hand side of (5). The conversions $\partial_z = (1/(2c))\partial_c$ and

$$\partial_{zz} = \frac{1}{4c^2}\partial_{cc} - \frac{1}{4c^3}\partial_c$$

give, after multiplication by $4c^3 F^{2-\alpha}$,

$$\begin{aligned}\partial_c(F^\alpha) F^{2-\alpha} &= 2\alpha(c + t)F, \\ \partial_{cc}(F^\alpha) F^{2-\alpha} &= 4\alpha(\alpha - 1)(c + t)^2 + 2\alpha F.\end{aligned}$$

The multiplied operator is the quadratic polynomial

$$\begin{aligned}E &= (1 - c^2)(c^3 \partial_{cc}^{\text{cl}} - c^2 \partial_c^{\text{cl}}) \\ &\quad + 2c^2[\lambda + 1 - (1 - 2\alpha - \lambda)c^2] \partial_c^{\text{cl}} - 4\alpha(\alpha + \lambda)c^3 F^2,\end{aligned}\tag{11}$$

where the cleared derivatives are the two displays above. Their coefficients in t are

$$\begin{aligned}(c + t)F &= 2ct^2 + (1 + 3c^2)t + c(1 + c^2), \\ \partial_c^{\text{cl}} &= 2\alpha(2ct^2 + (1 + 3c^2)t + c(1 + c^2)), \\ \partial_{cc}^{\text{cl}} &= 4\alpha(\alpha - 1)t^2 + 4\alpha c(2\alpha - 1)t + 2\alpha(1 + (2\alpha - 1)c^2).\end{aligned}$$

The linear coefficient of $\partial_{cc}^{\text{cl}}$ is $8\alpha(\alpha - 1)c + 4\alpha c = 4\alpha c(2\alpha - 1)$, and its constant term is $4\alpha(\alpha - 1)c^2 + 2\alpha(1 + c^2) = 2\alpha(1 + (2\alpha - 1)c^2)$. The square $F^2 = (1 + c^2 + 2ct)^2$ has coefficients $4c^2$, $4c(1 + c^2)$ and $(1 + c^2)^2$.

Coefficient of t^2 . The combination $c^3 \partial_{cc}^{\text{cl}} - c^2 \partial_c^{\text{cl}}$ has t^2 -coefficient $4\alpha c^3(\alpha - 1) - 4\alpha c^3 = 4\alpha c^3(\alpha - 2)$. Multiplication by $1 - c^2$ contributes $4\alpha c^3(\alpha - 2)(1 - c^2)$. The middle summand of (11) contributes $8\alpha c^3[\lambda + 1 - (1 - 2\alpha - \lambda)c^2]$, and the last summand contributes $-16\alpha(\alpha + \lambda)c^5$. Dividing the sum by $4\alpha c^3$ produces

$$(\alpha - 2)(1 - c^2) + 2[\lambda + 1 - (1 - 2\alpha - \lambda)c^2] - 4(\alpha + \lambda)c^2.$$

The terms independent of c^2 are $\alpha - 2 + 2\lambda + 2 = \alpha + 2\lambda$. The coefficient of c^2 is

$$-(\alpha - 2) - 2(1 - 2\alpha - \lambda) - 4(\alpha + \lambda) = -\alpha + 2 - 2 + 4\alpha + 2\lambda - 4\alpha - 4\lambda = -(\alpha + 2\lambda).$$

The quotient equals $(\alpha + 2\lambda)(1 - c^2)$, so the coefficient of t^2 in E is $-4\alpha c^3(\alpha + 2\lambda)(c^2 - 1)$.

Coefficient of t . The three contributions are

$$\begin{aligned} & (1 - c^2)(4\alpha(2\alpha - 1)c^4 - 2\alpha c^2(1 + 3c^2)), \\ & 4\alpha c^2(1 + 3c^2)[\lambda + 1 - (1 - 2\alpha - \lambda)c^2], \\ & -16\alpha(\alpha + \lambda)c^4(1 + c^2). \end{aligned}$$

Dividing by $2\alpha c^2$ produces

$$(1 - c^2)((4\alpha - 5)c^2 - 1) + 2(1 + 3c^2)[\lambda + 1 - (1 - 2\alpha - \lambda)c^2] - 8(\alpha + \lambda)c^2(1 + c^2).$$

The first summand equals $-(4\alpha - 5)c^4 + 4(\alpha - 1)c^2 - 1$. The second equals $2(\lambda + 1) + 6(\lambda + 1)c^2 - 2(1 - 2\alpha - \lambda)c^2 - 6(1 - 2\alpha - \lambda)c^4$. The third equals $-8(\alpha + \lambda)c^2 - 8(\alpha + \lambda)c^4$. The coefficient of c^4 in the sum is

$$\begin{aligned} & -(4\alpha - 5) - 6(1 - 2\alpha - \lambda) - 8(\alpha + \lambda) \\ & = -4\alpha + 5 - 6 + 12\alpha + 6\lambda - 8\alpha - 8\lambda = -(2\lambda + 1). \end{aligned}$$

The coefficient of c^2 is

$$\begin{aligned} & 4(\alpha - 1) + 6(\lambda + 1) - 2(1 - 2\alpha - \lambda) - 8(\alpha + \lambda) \\ & = 4\alpha - 4 + 6\lambda + 6 - 2 + 4\alpha + 2\lambda - 8\alpha - 8\lambda = 0. \end{aligned}$$

The remaining term is $-1 + 2(\lambda + 1) = 2\lambda + 1$. The quotient is therefore $(2\lambda + 1)(1 - c^4) = (2\lambda + 1)(1 - c^2)(1 + c^2)$, and the coefficient of t in E is $-2\alpha c^2(c^2 - 1)(c^2 + 1)(2\lambda + 1)$.

Constant term. The three contributions are $4\alpha(\alpha - 1)c^5(1 - c^2)$, $4\alpha c^3(1 + c^2)[\lambda + 1 - (1 - 2\alpha - \lambda)c^2]$ and $-4\alpha(\alpha + \lambda)c^3(1 + c^2)^2$. Dividing by $4\alpha c^3$ produces

$$(\alpha - 1)c^2(1 - c^2) + (1 + c^2)[\lambda + 1 - (1 - 2\alpha - \lambda)c^2] - (\alpha + \lambda)(1 + c^2)^2.$$

The coefficient of c^4 is $-(\alpha - 1) - (1 - 2\alpha - \lambda) - (\alpha + \lambda) = 0$. The coefficient of c^2 is

$$(\alpha - 1) + (\lambda + 1) - (1 - 2\alpha - \lambda) - 2(\alpha + \lambda) = \alpha - 1 + \lambda + 1 - 1 + 2\alpha + \lambda - 2\alpha - 2\lambda = \alpha - 1,$$

and the remaining term is $(\lambda + 1) - (\alpha + \lambda) = 1 - \alpha$. The quotient equals $(\alpha - 1)(c^2 - 1)$, so the constant term of E is $4\alpha c^3(\alpha - 1)(c^2 - 1)$.

These are the coefficients of $-2\alpha c^2(c^2 - 1)Q$. Both sides are quadratic polynomials in t , so $E = -2\alpha c^2(c^2 - 1)Q$. The divisions by α and by powers of c compare coefficients of polynomials and do not restrict the identity: both sides vanish to at least those orders. Dividing by $4c^3F^{2-\alpha}$ gives (5).

The initial slope

Start from (9), which is the differentiation of (6) at $z = 0$. With $t_\alpha = (\alpha + \lambda)/(\lambda + 1)$,

$$\begin{aligned} (1 - 2\alpha - \lambda - \alpha t_\alpha)(\lambda + 1) &= (1 - 2\alpha - \lambda)(\lambda + 1) - \alpha(\alpha + \lambda) \\ &= 1 - \lambda^2 - 2\alpha - 2\alpha\lambda - \alpha^2 - \alpha\lambda = M(\alpha), \end{aligned}$$

where $M(\alpha) = (1 - \lambda^2) - \alpha^2 - \alpha(3\lambda + 2)$. Thus $1 - 2\alpha - \lambda - \alpha t_\alpha = M(\alpha)/(\lambda + 1)$. With $\kappa t_\delta = t_\gamma$,

$$(1 - 2\gamma - \lambda - \gamma t_\gamma) - \kappa(1 - 2\delta - \lambda - \delta t_\delta) = \frac{M(\gamma)}{\lambda + 1} - \frac{\gamma + \lambda}{\delta + \lambda} \cdot \frac{M(\delta)}{\lambda + 1}.$$

The numerator of the common-denominator form is

$$(\delta + \lambda)M(\gamma) - (\gamma + \lambda)M(\delta).$$

The summand $(1 - \lambda^2)$ contributes $(\delta - \gamma)(1 - \lambda^2)$. The quadratic summand contributes

$$\begin{aligned} -(\delta + \lambda)\gamma^2 + (\gamma + \lambda)\delta^2 &= \gamma\delta(\delta - \gamma) + \lambda(\delta^2 - \gamma^2) \\ &= (\delta - \gamma)(\gamma\delta + \lambda(\gamma + \delta)). \end{aligned}$$

The remaining summand contributes

$$(3\lambda + 2)(-(\delta + \lambda)\gamma + (\gamma + \lambda)\delta) = (3\lambda + 2)(-\lambda\gamma + \lambda\delta) = \lambda(3\lambda + 2)(\delta - \gamma).$$

These are the three contributions

$$(1 - \lambda^2)(\delta - \gamma), \quad (\delta - \gamma)(\gamma\delta + \lambda(\gamma + \delta)), \quad \lambda(3\lambda + 2)(\delta - \gamma).$$

Their sum is

$$(\delta - \gamma)(1 + 2\lambda + 2\lambda^2 + \gamma\delta + \lambda(\gamma + \delta)) = (\delta - \gamma)((\gamma + \lambda)(\delta + \lambda) + (\lambda + 1)^2).$$

Dividing by $(\delta + \lambda)(\lambda + 1)$ and multiplying by $t_\gamma/(\lambda + 2)$ gives

$$\tau'_\gamma(0) - \psi'(0) = \frac{(\delta - \gamma)(\gamma + \lambda)((\gamma + \lambda)(\delta + \lambda) + (\lambda + 1)^2)}{(\delta + \lambda)(\lambda + 1)^2(\lambda + 2)}.$$

Coefficients of the contact polynomial

Write $P(t) = N_\gamma(t)(1 - \kappa z) - N_\psi(t)(1 - z)$ and

$$S = \frac{z(\delta - \gamma)}{\delta + \lambda}.$$

The constant terms contribute

$$(\gamma + \lambda)((1 - \kappa z) - (1 - z)) = (\gamma + \lambda)z(1 - \kappa) = S(\gamma + \lambda),$$

since $1 - \kappa = (\delta - \gamma)/(\delta + \lambda)$.

The coefficients of t^2 contribute

$$\begin{aligned} &-z(1 - z)\gamma(1 - \kappa z) + z(1 - \kappa z)\delta(1 - z) \\ &= z(1 - z)(1 - \kappa z)(\delta - \gamma). \end{aligned}$$

The relation $1 - \kappa z = (\delta + \lambda - z(\gamma + \lambda))/(\delta + \lambda)$ rewrites the right-hand side as SA_Q .

The coefficient of t in N_γ is $-\lambda + 1 - (1 - 2\gamma - \lambda)z$, and the coefficient of t in N_ψ is $-\lambda + 1 - (1 - 2\delta - \lambda)\kappa z$. Write $A = \lambda + 1$, $G = 1 - 2\gamma - \lambda$ and $D = 1 - 2\delta - \lambda$. The coefficient of t in P is

$$\begin{aligned} &-(A - Gz)(1 - \kappa z) + (A - D\kappa z)(1 - z) \\ &= -A + A\kappa z + Gz(1 - \kappa z) + A(1 - z) - D\kappa z(1 - z) \\ &= z(A(\kappa - 1) + G(1 - \kappa z) - \kappa D(1 - z)) = zB, \end{aligned}$$

where

$$B = (\lambda + 1)(\kappa - 1) + (1 - 2\gamma - \lambda)(1 - \kappa z) - \kappa(1 - 2\delta - \lambda)(1 - z).$$

Multiply by $\delta + \lambda$ and use $\kappa(\delta + \lambda) = \gamma + \lambda$:

$$\begin{aligned} (\delta + \lambda)B &= (\lambda + 1)(\gamma - \delta) + (\delta + \lambda)(1 - 2\gamma - \lambda)(1 - \kappa z) \\ &\quad - (\gamma + \lambda)(1 - 2\delta - \lambda)(1 - z). \end{aligned}$$

The right-hand side is affine in z . Its constant term is

$$\begin{aligned} K &= (\lambda + 1)(\gamma - \delta) + (\delta + \lambda)(1 - 2\gamma - \lambda) - (\gamma + \lambda)(1 - 2\delta - \lambda) \\ &= \lambda\gamma - \lambda\delta + \gamma - \delta + \delta - 2\gamma\delta - \lambda\delta + \lambda - 2\gamma\lambda - \lambda^2 \\ &\quad - \gamma + 2\gamma\delta + \gamma\lambda - \lambda + 2\delta\lambda + \lambda^2. \end{aligned}$$

The pairs $\lambda\gamma + \gamma\lambda - 2\gamma\lambda$, $-\lambda\delta - \lambda\delta + 2\delta\lambda$, $-2\gamma\delta + 2\gamma\delta$, $-\lambda^2 + \lambda^2$, $\lambda - \lambda$, $\gamma - \gamma$ and $-\delta + \delta$ each sum to zero, so $K = 0$.

The coefficient of z receives $-\kappa z$ from the second summand and $-(1 - z)$ from the third:

$$\begin{aligned} & -(\delta + \lambda)(1 - 2\gamma - \lambda)\kappa + (\gamma + \lambda)(1 - 2\delta - \lambda) \\ &= (\gamma + \lambda)(-(1 - 2\gamma - \lambda) + (1 - 2\delta - \lambda)) \\ &= 2(\gamma + \lambda)(\gamma - \delta). \end{aligned}$$

Hence $(\delta + \lambda)B = 2z(\gamma + \lambda)(\gamma - \delta)$, and $zB = -2z(\gamma + \lambda)S$. Adding the three contributions yields $P(t) = SQ(t)$.

This is an AI-generated solution to a problem that was prompted by Paata Ivanisvili in Cursor with Grok 4.7.

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